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Albert-Ludwigs-Universität Freiburg –
Department of Psychology
Chair of Nature Conservation and Landscape Ecology
Scandinavian Literature and Cultural Studies

Key Research Area "Languages of Knowledge"
Proposal: Networking Grant for Early-Career Researchers (2026)
Key Research Area Board
Prof. Dr. Eva von Contzen
Prof. Dr. Johanna Pink

15.01.2026

Project: Forests of Knowledge: Perceptions of the Forest in Literature Over Time

Dear Prof. Dr. Eva von Contzen and Prof. Dr. Johanna Pink,

on behalf of our interdisciplinary team, we are pleased to submit this proposal for the Networking Grant for Early-Career Researchers.

1. Project Title

Forests of Knowledge: Perceptions of the Forest in Literature Over Time

2. Short Description

This project investigates the forest as an epistemic and symbolic entity, analyzing how its narrative and normative framing in literature evolves across history. We utilize Large Language Models (LLMs) and Natural Language Processing (NLP) to conduct a diachronic analysis of a machine-readable corpus, identifying shifts in cultural perception and ecological knowledge over time. The project maps onto the 'Languages of Knowledge' fields by exploring forests as meaning-generating narrative spaces, identifying their long-term temporal perception (threat vs. sanctuary), and revealing how normative practices regarding the environment are culturally embedded (e.g., nature protection and exploitation).

3. Applicants and Idea Providers

- **Dr. Julius Fenn** (Department of Psychology): Expertise in Computational Modeling, Statistics, and Natural Language Processing (NLP); responsible for the technical execution and quantitative analysis.
- **Dr. Finn Rehling** (Chair of Nature Conservation and Landscape Ecology): Expertise in forest ecosystems and biodiversity; providing the ecological framework and environmental contextualization.

Albert-Ludwigs-Universität Freiburg

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- **Juliane Amberger** (Department of Scandinavian Studies): Expertise in narrative theory and cultural and literary history; overseeing corpus selection and qualitative literary interpretation.

4. Project Objectives and Methodology, Technical Implementation

This project investigates the forest as an epistemic and symbolic entity, analyzing

- **Methodological Foundation:** Develop a shared, open, and machine-readable literary corpus.
- **Scientific Inquiry:** Analyze temporal orders and normative understandings of forests, potentially including an analysis of "biodiversity loss" in literary descriptions over time.
- **Network Building:** Establish a stable interdisciplinary research network and prepare a major third-party funding proposal.

Methodology. The analysis utilizes NLP/ LLMs to capture diachronic trends. Following the approach of Foramitti et al. (2025) for analyzing long-term linguistic and emotional shifts, we are planning to conduct the following analyses:

- **NLP/LLM Framework:** Capturing diachronic trends via vector embeddings and prompt-based learning.
- **Retrieval-Augmented Generation (RAG) Framework:** Grounding LLM interpretations in specific retrieved passages to ensure corpus-based accuracy.
- **Sentiment & Framing:** Quantifying emotional valence (threat vs. sanctuary) and classifying themes (Utility, Recreation, Imagination).
- **Trend Analysis:** Using statistical modeling to identify historical breaks and semantic continuities.

Technical Implementation. The infrastructure leverages automated data acquisition and vector-indexed storage to manage large-scale literary corpora, we are planning to implement the following technical components:

- **Data Sourcing:** Automated collection from literary databases like Project Gutenberg using Selenium-based web scraping.
- **Storage:** Implementation of a vector database (Supabase) for text and metadata.
- **Processing:** Generating embeddings via the Qwen model and classification via BERT-based architectures.
- **Interface:** Deployment of a web-based UI via the University of Freiburg Rechenzentrum.

5. Expected Innovation

This project addresses a critical research gap at the intersection of literary studies and computational science by introducing AI-driven, diachronic analysis to a field currently dominated by synchronic or qualitative studies and traditional statistical approaches. While existing Digital Humanities resources provide extensive

corpora, they often lack machine-driven workflows necessary to scale a "Wide Reading" approach to an unprecedented level, allowing us to objectify "forest knowledge" across centuries. This unique synergy between environmental humanities and computational statistics enables us to quantify long-term shifts in symbolic, narrative, and normative forest perceptions—transforming intertextual analysis from manual close reading into a verifiable, large-scale mapping of cultural and ecological change. Beyond the immediate project scope, the resulting infrastructure—including the machine-readable corpus and standardized annotation guidelines—will serve as a long-term foundation for further research at the university, providing a methodology for analyzing cultural perceptions of nature that is directly transferable to other ecological or environmental topics.

6. Work Packages (WP), Ressource Planning

- **WP1:** Consolidation of terminology, concepts, and corpus design.
- **WP2:** Automated scraping and initial database setup.
- **WP3:** Manual curation, annotation, and classification by a student assistant.
- **WP4:** Conducting university wide expert workshop and start writing of a grant proposal.

Personnel Costs: We request €3,000.00 for a graduate student assistant (WHK) to support WP3. The assistant's primary responsibility is the manual coding and curation of the text corpus to ensure high-quality data. All software development, programming, and technical infrastructure (RAG and database setup) will be handled directly by the applicants. This funding covers approximately 14–16 hours/month in 2026.

Sincerely the Project Team,

(Julius Fenn, postdoctoral researcher at the Cluster of Excellence Living, Adaptive and Energy-autonomous Materials Systems)

(Finn Rehling, scientific manager of MultiTroph, Chair of Nature conservation and Landscape ecology, Faculty of Environment and Natural resources)

(Juliane Amberger, doctoral researcher at the Scandinavian Department)